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B.Tech. Degree VII Semester Special Supplementary Examination March 2023

ME 19-205-0702 VIBRATION AND NOISE CONTROL (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Model a given physical system using spring, mass, and damping elements.

CO2: Formulate the governing differential equations of single dof, multi dof, and continuous systems.

CO3: Solve the governing equations and interpret the results.

CO4: Understand the effects of noise and the control measures adopted.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

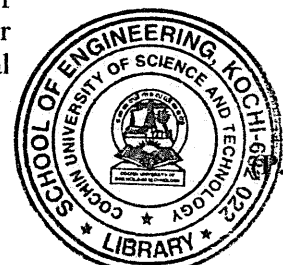
PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) A spring-mass system (k_1 - m_1) has a natural frequency f_1 . Calculate the value of k_2 another spring which when connected to k_1 in parallel increases the frequency by 30 percentage.	3	3	L2	1	1
(b) A torsional pendulum when immersed in oil indicates its natural frequency as 200 Hz but when it was put to vibration in vacuum having no damping its natural frequency was 250 Hz. Find the value of damping factor of oil.	3	3	L3	1	2
(c) Find the ratio of amplitude of torsional vibration for a two rotor system.	3	3	L1	4	1
(d) Explain the theory behind the working of accelerometer and seismometer.	3	3	L2	4	2
(e) Apply Rayleigh's method to find the natural frequency of transverse vibration of shaft carrying three point loads.	3	3	L3	3	1
(f) What do you mean by a continuous system?	3	3	L1	4	2
(g) Define continuous and intermittent noise.	3	3	L1	4	3
(h) Explain Octave band analysis and its importance.	3	3	L2	4	2

PART B

(4 × 12 = 48)

- II. (a) A slab door, 2 m high, 0.75 m wide, 40 mm thick, and with a mass of 36 kg, is fitted with an automatic door closer. The door opens against a torsion spring with a modulus of 10 N-m/radian. Determine the necessary damping to critically dampen the return swing of the door. If the door is opened 90 degree and released, how long will it take until the door is within 1 degree of closing?
- (b) A horizontal spring mass system with Coulomb damping has a mass of 5 kg attached to a spring of stiffness 980 N/m. If the coefficient of friction is 0.025, calculate the frequency of free vibrations, the number of cycles corresponding to 50% reduction in amplitude if the initial amplitude is 5cm and the time taken to achieve this 50% reduction.

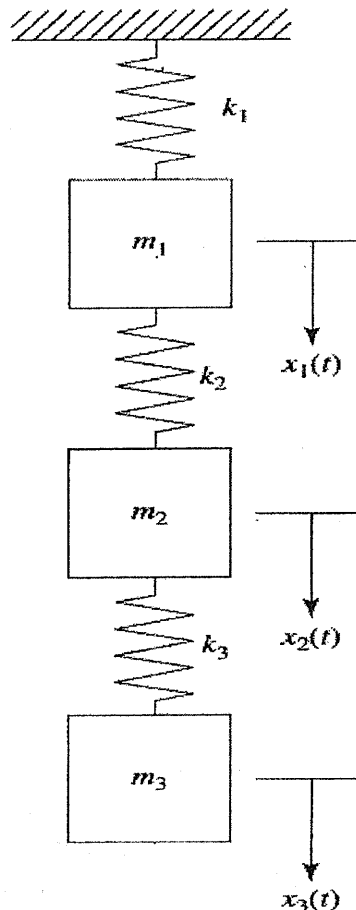
OR

T.O.)

BTS-VII(S.S)-03-23-1476

	Marks	BL	CO	PO
III. (a) A single cylinder vertical petrol engine of total mass 320 kg is mounted upon a steel chassis and causes a vertical static deflection of 2 mm. The reciprocating parts of the engine have a mass (m) of 24 kg and moving through a vertical stroke of 150 mm with simple harmonic motion. A dashpot attached to the system offers a resistance damping force of 490 N at a velocity of 0.3 m/sec. Determine: (i) The amplitude of steady state vibration when the driving shaft of the engine rotates at 480 rpm (ii) The speed of the driving shaft (in rpm) at resonance.	6	L3	1	2
(b) The vibration of the platform of railway station are periodic at the frequency range of 12-50 Hz. A vibration measuring instrument is to be installed on some foundation independent of the platform. The small foundation is supported by four identical springs resting on the platform. The total mass of the instrument and foundation is 50 Kg. What is the maximum value of spring stiffness if the amplitude of transmitted vibration is to be less than 10% of the platforms vibration over the given frequency range. Take ($\epsilon = 0.20$. System is treated as single degree of freedom).	6	L3	1	2

- IV. Determine the natural frequencies and mode shapes of the spring mass system shown in figure, where $K_1 = 2K$, $K_2 = K$, $K_3 = K$ and $m_1 = 2m$, $m_2 = 2m$, $m_3 = m$.



OR

(Continued)

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		Marks	BL	CO	PO
V.	A 4-cylinder engine coupled to a propeller are approximated to a three rotor system in which the engine is equivalent to a rotor of moment of inertia 800 kgm^2 the flywheel second rotor 320 kgm^2 and the propeller to a third rotor 20 kgm^2 . The first and second rotors being connected by 50 mm diameter 2 meter long shaft and the second and third rotors connected by 25 mm diameter and 2 meter long shaft. Neglecting the inertia of the shaft taking the modulus of rigidity 80 GN/m^2 , determine: (i) Natural frequency of torsional oscillations (ii) The position of nodes.	12	L3	2	2
VI.	A shaft 1.5 m long, supported in flexible bearings at the ends carries two wheels each of 50 kg mass. One wheel is situated at the centre of the shaft and the other at a distance of 375 mm from the centre towards left. The shaft is hollow of external diameter 75 mm and internal diameter 40 mm. The density of the shaft material is 7700 kg/m^3 and its modulus of elasticity is 200 GN/m^2 . Find the approximate fundamental frequency of vibration of the system.	12	L3	3	2
OR					
VII.	A beam of uniform cross-section having length l is fixed at both ends. Derive the expression for the frequency of transverse free vibration of the beam.	12	L3	3	2
VIII.	(a) An operator in a textile mill is operation five machines. The sound pressure levels of the machines at his positions are 95, 90, 92, 88 and 82 dB respectively. (i) What is the total sound pressure level at his position due to all the five machines? (ii) Determine the sound pressure level at operator's position if machine 2 and 3 are turned off.	6	L3	4	1
	(b) Show that as the distance from a point source doubles, the sound intensity level decreases by 6dB.	6	L3	4	1
OR					
IX.	Explain: (i) Noise control at the source. (ii) Noise control along path. (iii) Noise control at the receiver.	12	L2	4	1

Bloom's Taxonomy Levels

L1 = 15.8 %, L2 = 21.1%, L3 = 63.1%
